Manual

Energy Management - Manual Input

EMG-AME 8.3

Version 1.0.23133

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Copyright

Contents

1 Energy Management - Manual Input

Purpose

Planning, collection and manual collection of energy data.

You use the function package for the following purposes:

- You want to collect energy counters manually on the MOC.

Integration

You require the function package EMG-MGM for the counter readings and the manual input.

If you want to collect data via HYDRA-SMA, you require the SMA package for mobile data collection. The data collection via SMA is not part of this license.

Features

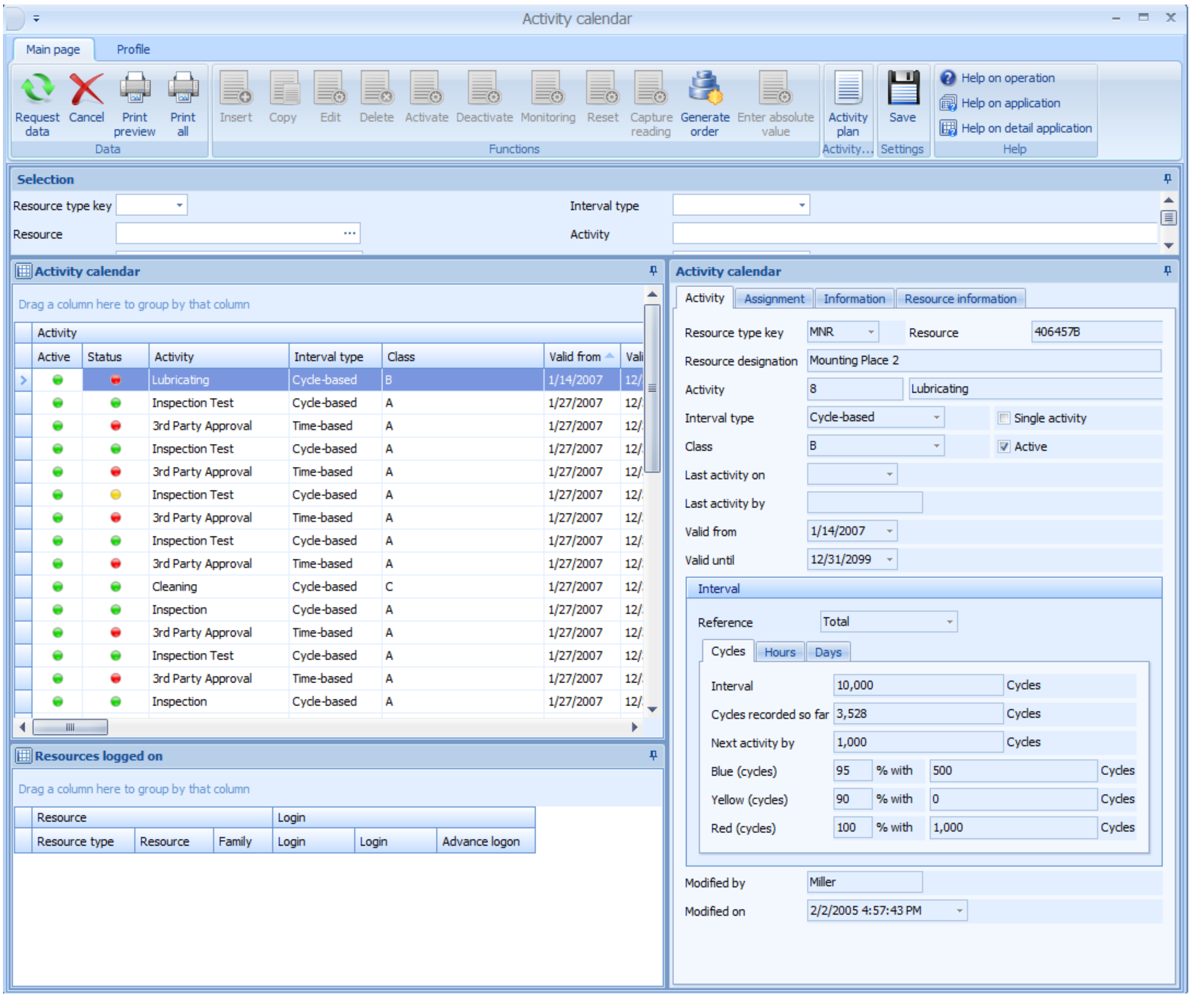
- Input plans
    - Definition of input plans for the different counter resources that must be collected manually.
    - Definition of a sequence for counter reading processes that are performed on the same day. The sequence then also specifies the route.
- Data collection
    - Collection of counter values via input function of the counter reading plan (MOC)

2 Maintenance Calendar (Activity Calendar)

Overview

Menu	Resource Management → Current information → Maintenance calendar
Transaction code	rmcal
Function authorization	rmcal

This overview is a valuable aid for supervisors and maintenance personnel because only through regular maintenance, the expensive resources (e.g. production machines or tools) can retain their production quality and do not cause unnecessary breakdowns. You can also use the calendar as basis for maintenance planning and as data collection tool to record the activities performed.



Purpose

The maintenance calendar or activity calendar has been designed to plan and show maintenance activities or other recurring activities. Activities can be maintenance, test equipment calibration, reading of energy counters and so on. Use the field *Activity type* to identify the relevant function and type of activity. In most cases, this field remains empty which means that a maintenance or similar activity is scheduled for a resource. For special requirements, e.g. calibration of test equipment, enter the relevant identifier in this field ("K" to identify calibration of test equipment). That means, the user can differentiate between the type of activity.

This document describes how to use the calendar in the HYDRA Tool and Resource Management, Energy Management and Gage Calibration.

The task of maintenance and activity monitoring is to track the configured activities and perform the following actions:

Refreshing the current values
- "Cycles recorded so far" (cycle-based maintenance/activity) or
- "Hours recorded so far" (maintenance/activity via hours of operation).

The status of an activity is set when a configured threshold value (blue/yellow/red) has been exceeded, and this event is documented in the database (to generate evaluations via the resource history). The threshold values are checked in the following order: red > yellow > blue. This means, the system checks first if the threshold value "red" has been exceeded. If so, this status is set and documented. Otherwise, the inspection is continued for the threshold values "yellow" and then "blue".

Monitoring is only run for activities that are marked active and whose validity period includes the current time.

For this purpose, the monitoring process hywtkupd.out/.exe is embedded in the HYDRA scheduler and cyclically called. .

You can define any number of activities for each resource. Several activities can thereby be defined for each resource. The following types of intervals are available when defining the maintenance times:

Cycle-based activity

The system compares target and actual cycles with a cycle-based activity. The difference between the two values indicates when maintenance is due. The actual cycles are automatically recorded in HYDRA.

Requirement: a cycle monitoring must be performed for the machine.

Activity based on hours of operation

The times recorded in HYDRA are used for a maintenance/activity based on hours of operation. In the *Resource type*, you specify the resource performance accounts that are used to calculate the hours of operation. The activity is due when the interval defined in the maintenance calendar has been reached.

Time-based activity (days)

With this type of activity, the system calculates the next maintenance due date using the number of days specified for this activity. This number of days is based on the Gregorian calendar.

Single activity

Combined with the above-mentioned types of intervals, you can even specify an activity as "non-recurring". After the reset, the activity is deactivated automatically.

Additional notes:

When data is selected, the user can only view the activities of resources that are included in the responsibility area the user is authorized for.

You must have defined the resource master data in the resource stock, which is required for the maintenance calendar.

To use resources with cycle-based activities and maintenances based on hours of operation, it must be configured in the system that you can post to these resources. This can be configured via the resource type.

For resources with DNC processing, no cycle-based activities and no activities based on hours of operation can be defined, as you cannot post data for these resources.

And also for energy counters this is not possible, as energy counters do not use machine cycles as data basis.

Selection criteria

The application provides the following selection criteria:

Resource type

Selection of the specified resource type.

Resource

Selection of the specified resource.

Field description of the *Activity* tab

Resource type

Shows the resource type for the defined the activity.

To use resources with cycle-based activities and maintenances based on hours of operation, it must be configured in the system that you can post to these resources. This can be configured via the resource type.

For resource types with DNC processing, no cycle-based maintenance and no maintenance based on hours of operation can be defined, as you cannot post data for these resources.

Resource

Shows the resource the activity is defined for.

Activity

Description of the activity. When you create an activity, the system automatically allocates a number to the name and this number identifies the activity.

Type

Select a type to specify how monitoring is carried out:

T	Cycle-based activity
B	Activity based on hours of operation
Z	Time-based activity

Depending on the above selection, one of the tabs *Cycles*, *Hours* or *Days* is released.

Single activity

If this option is set, the activity is only carried out once. In this case, the "interval" field is hidden. When reset, the activity is automatically deactivated.

Class

This input field is used to classify maintenance activities. For example, all cleaning activities can be classified as "Cleaning".

Using the grouping function in the overview screen, you can combine and display all maintenance activities that logically belong to the same class. Otherwise, this entry field is used for comments.

Active

It is possible to deactivate activities temporarily, and reactivate them again at a later time. The following display shows if an activity is currently active or not:

Activity activated
Activity deactivated

Deactivated activities are not integrated in the monitoring.

Authorization (as of service pack 16/2020)

You can use the authorization level to specify the maintenances in detail that a person is allowed to reset on the Windows terminal AIP.

A person can reset a maintenance if the following conditions are fulfilled:

- The option (checkbox) *Reset maintenances* in the HR master data is activated;

- An authorization level is entered in the HR master data;

- The authorization level of the person must be greater than or equal to the authorization level specified in field *Authorization* of the maintenance.

The system performs an online validation check on the HYDRA server. The system only performs the authorization check if a staff badge number has been entered in the AIP input dialog.

If no authorization level is specified, then the maintenance can always be reset and the authorization level stored in the HR master data is not relevant (downward compatibility).

Status

A colored signal in front of each activity clearly shows if a maintenance interval will soon expire or has already expired, and if a maintenance activity must be carried out. This way, the user can see at one glance which activity must be carried out or is already overdue.

The user specifies threshold values, which change the color of the signal. For cycle-based activities and maintenances based on hours of operation, the application shows percentages, for time-based activities days are shown. The following 4 colors can be defined (each corresponding to a status type):

green
blue
yellow
red

For further information, please refer to the descriptions of the different interval types (see below).

Note:

The status of the resource itself does not change when a maintenance status is reached.

Last activity

This shows the time when the maintenance activity was last reset and the name of the user who reset the maintenance.

Please note that the time of resetting the maintenance may deviate from the time that the maintenance has actually been carried out.

Valid from, valid until

You specify these times to assign time limits to maintenance activities.

If the current point in time is not included in the validity period of a maintenance activity, then this maintenance is not integrated in the maintenance monitoring, i.e. the maintenance status is not updated.

Modified by

Name of the last user who edited the maintenance activity and time of the last change.

The different tabs are explained in the following:

Field descriptions of the *Cycles* sub-tab

Interval

After the number of machine cycles specified in this field, the maintenance must be carried out. This number and the two values below refer to the value in the *Reference* field (see below).

This field is hidden if it is a singular/non-recurring maintenance.

Cycles recorded so far

Here, the system displays the number of machine cycles recorded so far in HYDRA. This value is updated by a cyclical process. For further information, refer to the section *Maintenance monitoring*.

Next activity

If you create a new activity, then this value is calculated by default using the current actual value (actual cycles) + the specified interval.

When you reset the maintenance activity, the system calculates this value and shows when the next activity is due.

When an activity is reset, you can use the *Calculation base* option to specify how the next maintenance due date is calculated:

Target value	The value for the next activity is based on the current value: Next activity after = current value (next activity after) plus interval
Actual value	The value for the next activity is based on the number of previously recorded cycles: Next activity after = cycles recorded so far + interval

Reference

For the above values, you must always consider this option. The following values are possible:

G	Total Activity monitoring is based on the total number of cycles recorded so far.
A	Relating to order/OP If a resource is logged on with the operation logon, then it is checked if activities with reference = A exist for this resource. These activities are then automatically reset. The *monitoring* now checks whether the interval has been reached using cycles posted for the operation currently logged on and sets the status accordingly.

Order-related monitoring is not available for resources of the type "MNR“ (machines)!!

This type of maintenance monitoring only makes sense for resources that are logged on to one operation at a given time. This means that a maximum of one operation may be logged on to the workstation/machine.

The posting of cycles to a resource does not take place in real time, but at longer intervals (e.g. at logoff, at interruption of an operation or during an automatic change of shifts). Therefore, this type of monitoring only makes sense for operations with a long runtime.

Blue / Yellow / Red

Enter the threshold values as percentages that identify the status of a maintenance activity.

"Cycles recorded so far" < Blue-% from "Next maintenance after"	green
"Cycles recorded so far" >= Blue-% and < Yellow-% from "Next maintenance after"	blue
"Cycles recorded so far" >= Yellow-% and < Red-% from "Next maintenance after"	yellow
"Cycles recorded so far" = Red-% from "Next maintenance after"	red

The relevant values specify the signal color.

Notes

The threshold values can be greater than 100%.

No validation check is made with regard to the order of the threshold values.

Field descriptions of the *Hours* sub-tab

Interval

Enter the period of time after which the maintenance activity must be run. This value, and the two following values, refer to the value in the *Reference* field (see below).

This field is hidden if this is a single activity.

Hours recorded so far

Here, HYDRA displays the time that has been posted for this resource so far. This value is updated by a cyclical process.

Please note that for the hours recorded so far, the RPA times are used that are identified as such in the *Resource type* (option *RPAs as hours of operation in the Maintenance calendar*).

Next activity after

This value is calculated by default from the current actual value (hours recorded so far) plus the specified interval.

When the activity is reset, this value is calculated and shows when the next maintenance activity is due.

When a maintenance activity is reset, you can specify how the next activity is to be calculated using the "Calculation base" option:

Target value	The value for the next activity is based on the current value: Next activity after = current value (next activity after) plus interval

Actual value The value for the next activity is based on the number of hours recorded so far:
Next activity after = hours recorded so far + interval

Reference

For the above values, you must always consider this option. The following values are possible:

G Total
Activity monitoring is based on the total time that has been posted so far.

A Relating to order/OP
If a resource is logged on with the operation logon, then it is checked if activities with reference = A exist for this resource. These activities are then automatically reset.

On the basis of the duration posted for the currently logged on operation, monitoring now checks whether the interval has been reached, and then sets the status accordingly.

This type of maintenance monitoring only makes sense for resources that are logged on to one operation at a given time. This means that a maximum of one operation may be logged on to the workstation/machine.

The posting of cycles to a resource does not take place in real time, but at longer intervals (e.g. at logoff, at interruption of an operation or during an automatic change of shifts). Therefore, this type of monitoring only makes sense for operations with a long runtime.

Blue / Yellow / Red

Enter the threshold values as percentages that identify the status of a maintenance activity.

"Hours recorded so far" < Blue-% from "Next activity after"	green
"Hours recorded so far" >= Blue-% and < Yellow-% from "Next activity after"	blue
"Hours recorded so far" >= Yellow-% and < Red-% from "Next activity after"	yellow
"Hours recorded so far" >= Red-% from "Next activity after"	red

The relevant values specify the signal color.

The threshold values can be greater than 100%.
No validation check is made with regard to the order of the threshold values.

Field descriptions of the sub-tab *Days*

Interval

Interval in days after which an activity is to be performed. The interval is based on the Gregorian calendar.

This field is hidden if this is a single activity.

Next activity on

Date when the next activity is due.

When a new maintenance is created, this value is calculated by default from the current date plus the specified interval.

Blue / Yellow / Red

Enter the number of days that specifies the status of a maintenance activity. The color of the signal is based on the remaining time, i.e. the difference between the date of the next activity and the current date ("today").

Remaining time <= "Red" value	red
Remaining time <= "Yellow" value	yellow
Remaining time <= "Blue" value	blue
Other	green

The threshold values can be greater than 100%.
No validation check is made with regard to the order of the threshold values.

Field description of the *Assignment* tab

Order

This field is only relevant in connection with the additional feature *Generate maintenance orders* or if you generate calibration (inspection) orders. Assign a maintenance order/calibration order using the "Create order" function that you call with this button .

If this field is filled, then the order number refers to a maintenance/calibration order. The activity will automatically be reset if the maintenance/calibration order is finished. When the maintenance/calibration order is finished for this activity, the order number is also removed from this input field.

Project number

This field is only relevant with activity type "K" (calibration). Depending on the system configuration, two different variants exist.

Variant 1 (there is exactly one work plan for all calibration inspection plans):

=> Input of the calibration inspection plan number (without version number)

Variant 2 (there is a separate work plan for each calibration inspection plan)

=> should remain empty. If you fill the field, the work plan list called in the application Generation of orders is pre-filtered by this project number.

Planned order

Control field that is currently not used. Remains empty.

Cost object

Control field that is currently not used. Remains empty.

Activity type

Identifies the activity type: For example, calibrations are identified by the type K.

Field descriptions of the *Information* tab

You can store a short description of the maintenance activity to ensure that the user or the maintenance worker receive more details on running the activity (e.g. notes on regulations to be observed, materials to be used).

Field descriptions of the *Resource information* tab

Inventory number

Shows the inventory number stored in the resource configuration. Additional information in form of comments.

Engraving number

Shows the engraving number on the device (machine, radiator etc.) stored in the resource configuration. Additional information in form of comments.

Drawing number

Shows the drawing number stored in the resource configuration. Additional information in form of comments.

Manufacturer

Shows the drawing number stored in the resource configuration. Additional information in form of comments.

Owner

Shows the owner name stored in the resource configuration. Additional information in form of comments.

Toolbar

Activate

Function authorization: rmcal.active

Opens the editing dialog to activate an activity

Deactivate

Function authorization: rmcal.deactive

Opens the editing dialog to deactivate an active activity

Monitoring

Function authorization: rmcal.monitor

Updates the status of the activities

Reset

Function authorization: rmcal.reset

Opens an editing dialog to reset an activity

Capture reading (EMG 8.1 only)

Function authorization: rmcal.captvalues

Opens the editing dialog to enter a counter reading. Any number of difference values may be entered within a time interval. Therefore, it is a delta collection. For this reason, it is also possible to subsequently enter data relating to the past. If you do not enter date values in the fields, the system generates the interval using the last data capture as start time and the current posting time as the end of the interval.

Enter absolute value (EMG 8.1 only)

Function authorization: rmcal.captvalues

Opens the editing dialog to enter a counter reading using an absolute value. The system calculates the difference to the previous reading. It is not possible to enter values for periods in the past. The system sets the start time of the interval to the end of the last entry. If an end is not entered, the system uses the current posting time.

Generate order

Function authorization: rmcal.generate

Creates an order that is used for organizational processing of the activity. Once the order is finished, you can set an option to have the activity automatically reset.

As of service pack 13/2018, you can generate orders automatically. Please find further details in section "Automatic generation of orders".

Activity plan (EMG 8.1 only)

Function authorization: rmcal.timetable

Opens the report Maintenance plan.

Document management (WRM-WWR 8.2 only)

Function authorization: rmcaldoc

Click this button to call the Document management.

Detail application *Resources logged on*

For resources of type "MNR", the detail application provides additional information on the resources currently logged on.

Resource type, resource

The currently selected resource and its resource type.

Resource, family, resource type

Resource that is logged on, the family and the resource type.

Login

Date and time of the resource login.

Advance logon

Identifier of the advance logon.

Automatic generation of orders

When a threshold value (blue, yellow or red) is reached for an entry in the activity calendar, an associated order can be generated automatically. In the INI configuration, you define the resource type and the threshold value to generate the order automatically. If the threshold value is exceeded, then the system does not generate an order for the exceeded threshold value. If no relevant INI configuration exists, then the automatic order generation is omitted.

INI configuration

Create the INI configuration with the name „MAINTENANCECALENDAR“ and the MOC user "0". Create a new entry in the INI configuration:

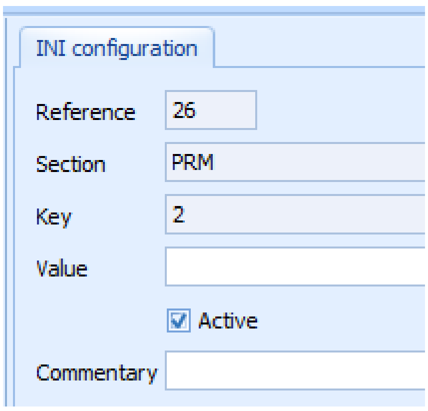
Field name	Value
Name	MAINTENANCECALENDAR
Description	Configuration maintenance/activity calendar

Field name	Value
MOC user	0

Create the required configurations for this INI configuration. For this entry, create an entry including the following values in the INI data configuration:

Field name	Value	Note
Section	<Resource type>	e.g. PRM (for test equipment)
Key	<threshold value>	Threshold value blue: 1 Threshold value yellow: 2 Threshold value read: 3
Value		Leave field empty
Active	Yes	

The below screenshot outlines a configuration to generate a calibration order for the resource type "PRM" (test equipment) when the threshold value 2 "yellow" is reached.



Order generation

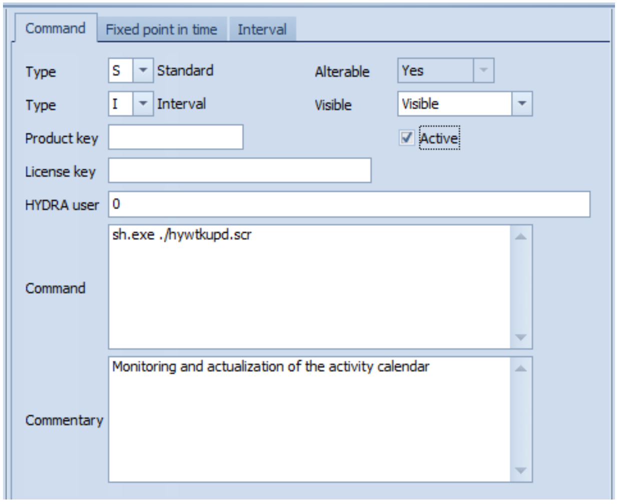
For automatic order generation it is assumed that for each resource type defined in the INI configuration in the MOC application (transaction code "edworgen" - separate licensing required) a corresponding configuration with matching object and assigned work plan is defined.

For automatic order generation it is assumed that for each resource type defined in the INI configuration in the MOC application „Hyperlink"(transaction code "edworgen" - separate licensing required) a corresponding configuration with matching object and assigned work plan is defined.

..\..\functions\MOC\MOC_OrderAutomaticGeneration.pdf

Scheduler job

Another requirement is that the scheduler job hywtkupd.scr is active that is responsible for the maintenance status.



Create calibration order

To create a calibration order, the system also requires that the calibration inspection plan is stored in the field *Project number* of the activity calendar.

The program, which is called by the scheduler job, only generates an order if a configured threshold is actually exceeded. If a threshold has already been exceeded and the scheduler job is started again, then the program does not generate an order for this threshold.

The program also does not generate an order if the threshold 2 is configured, for example, and it is now changed from threshold 1 directly to threshold 3 because of the actual data.

Report *Maintenance plan* (EMG 8.1 only)

The report *Maintenance plan* shows all maintenance activities that are displayed in the calling application *Activity calendar*.

The maintenance activities are displayed in the following sequence:

- descending, sorted by state
- ascending, sorted by priority

The following data is displayed for each maintenance activity in the maintenance plan:

<u>Left-hand column</u>

- State (0: green, 1 blue, 2 yellow, 3 red)
- Maintenance type

- Description
- Resource type
- Resource

Central column

- Next maintenance after (cycles)
- Cycles recorded so far
- Next maintenance after (hours)
- Hours recorded so far
- Next maintenance on

Right-hand column

- Information